STATISTICS

Chapter 6 Homework

Normal distributions and percentiles

Sections 6.1-6.2 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Location, area, and standardization

1. Package masses are normally distributed with mean 500 g and SD 10 g. Sketch the curve and label the mean and positions one, two, and three SDs from the mean.

2. For Problem 1, find z-scores for 485 g and 520 g. Explain the sign and magnitude of each. Is a z-score itself a probability?

3. Use the empirical rule to estimate the probability that a package weighs between 480 g and 520 g. Estimate the probability above 530 g.

4. Find $P(X < 485)$, $P(X > 520)$, and $P(485 < X < 520)$, where X is package mass. Show the z conversion and shade each region. Use the cumulative values below.

z	-1.50	0	2.00
$P(Z \le z)$	0.0668	0.5000	0.9772

5. Find $P(|X - 500| > 20)$ for the masses in Problem 1. Explain why two tails are needed. Use $P(Z \le 2) = 0.9772$ and symmetry.

Reverse questions and comparisons

6. Find the 90th percentile of package mass using $z = 1.2816$. Interpret the percentile in context.

7. Find the central 95% interval of individual package masses using $z = \pm 1.96$. Explain why this is not an interval estimating an unknown population mean.

8. Machine B produces masses with mean 800 g and SD 20 g, also normally distributed. Which is more unusually high: 515 g from the first machine or 820 g from B? Compare standardized distances.

9. Given that a package from the first machine weighs more than 500 g, find the probability it exceeds 520 g. Use the tail values from Problem 4 and state the new denominator.

10. A wait-time dataset is strongly right-skewed but has a mean and SD. A student standardizes one observation and immediately uses a normal table. Explain the missing assumption and why standardizing does not supply it.

11. Suppose the first machine's mean stays 500 g while its SD drops to 5 g. Without finding a new table value, explain how $P(X > 520)$ changes. Compute the new z-score.